The Glucose Mystery

The puzzle: We know the molecular formula is $C_6H_{12}O_6$, but what is the exact spatial arrangement? To crack the case, chemists interrogated the molecule using 6 classic reactions.

The Detective Work Starts Here!

Clue #1: The Straight Chain

$$\text{Glucose} \xrightarrow{\text{Prolonged heating with HI } \triangle} \text{n-hexane}$$

Reaction alert! Proof of the backbone!

Deduction: All six carbon atoms are linked in a straight, unbranched chain.

Clues #2 & #3: The Carbonyl Group

Glucose + NH_2OH
(Hydroxylamine)

↓

Oxime

Glucose + HCN
(Hydrogen cyanide)

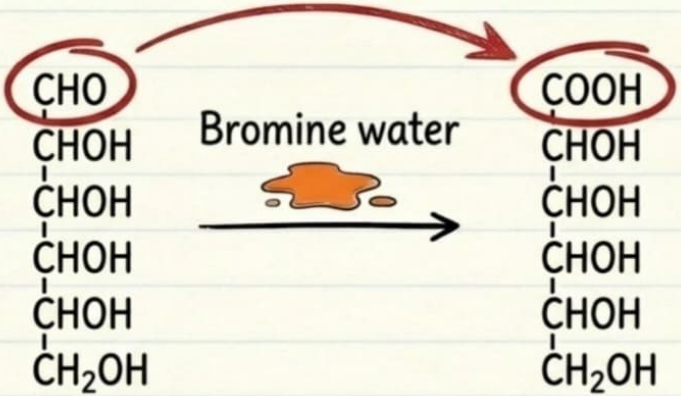
↓

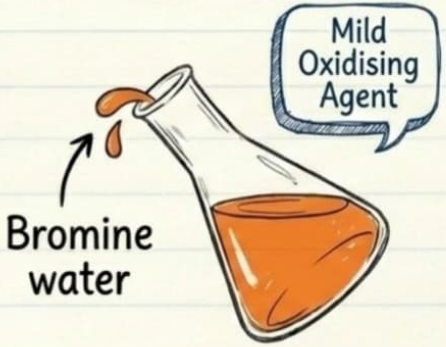
Cyanohydrin

Simplified:
If it reacts with these two, a >C=O is definitely present.

Deduction: Both reactions confirm the presence of a carbonyl group (>C=O).

Clue #4: Identifying the Aldehyde



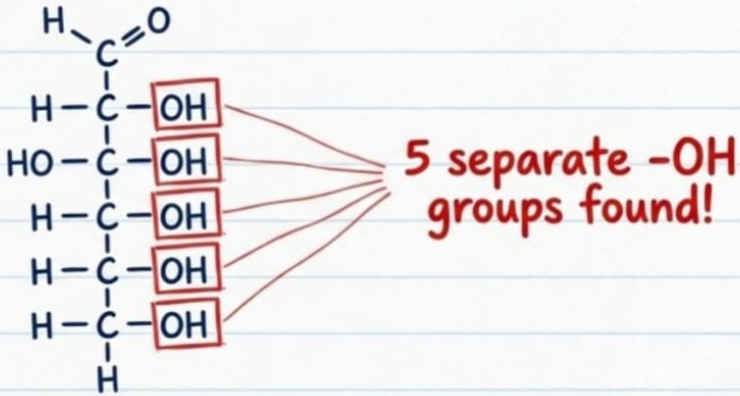


Glucose + Bromine water → Gluconic Acid (a 6-carbon carboxylic acid).

Deduction: The carbonyl group must be an aldehydic group at the end of the chain.

Key takeaway: Ketones don't oxidise this easily! Must be an aldehyde.

Clue #5: Counting the Hydroxyls



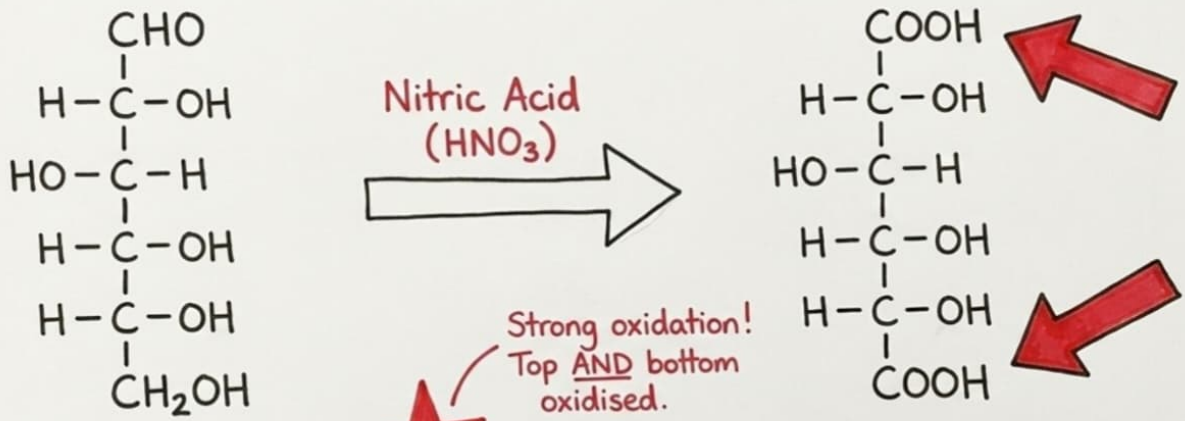
Glucose + Acetic anhydride ⟶ Glucose pentaacetate.

Penta = 5!

Deduction: Confirms the presence of EXACTLY five -OH groups. Because the resulting compound is stable, these five groups must be attached to different carbon atoms.

Stable structure = separate carbons.

Clue #6: Finding the Primary Alcohol

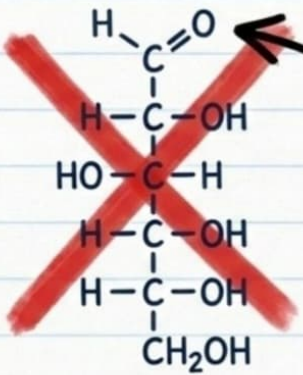


Glucose (or Gluconic acid) + Nitric Acid (HNO_3) → Saccharic Acid (a dicarboxylic acid).

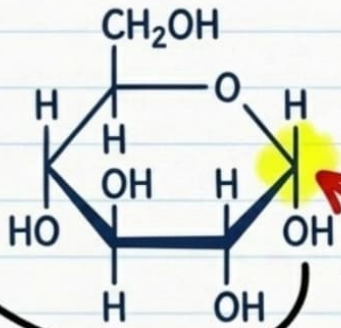
Deduction: Confirms the presence of a primary alcoholic group ($-CH_2OH$) at the very bottom of the chain.

The Plot Twist! (The Cyclic Structure)

Myth



Reality



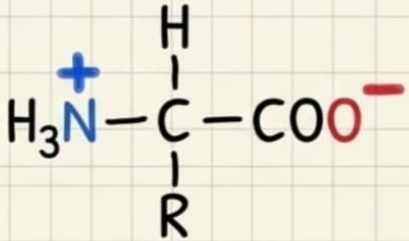
The Anomeric Carbon is born!

The Problem: Wait! The open chain doesn't explain everything. Glucose doesn't give a Schiff's test, and it exists in two distinct crystalline forms (α and β).

The Solution: The 5th -OH group attacks the aldehyde to form a 6-membered cyclic hemiacetal ring (Pyranose).

Chapter 2: The Building Blocks (Amino Acids)

Core Concept: Amino acids contain both an amino group ($-NH_2$) and a carboxyl group ($-COOH$).

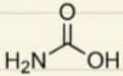


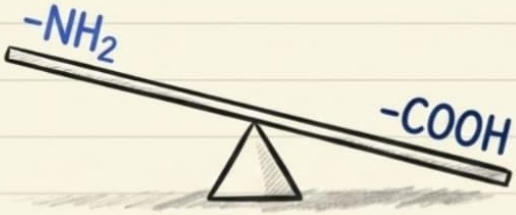
The **Zwitterion:** In aqueous solution, the carboxyl group loses a proton and the amino group accepts one. It becomes a neutral, dipolar ion.

Zwitterion: Neutral overall but carries charges! Gives them Amphoteric behaviour (reacts with acids AND bases).

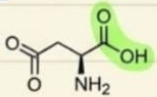
Classification: The Chemical Math

Neutral:
$-NH_2$ count = $-COOH$ count
(e.g., Glycine).

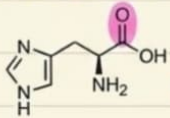




Acidic:
$-COOH$ count > $-NH_2$ count
(e.g., Aspartic acid).



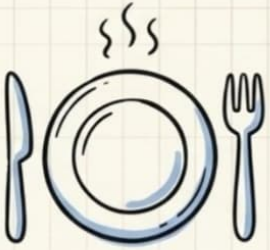
Basic:
$-NH_2$ count > $-COOH$ count
(e.g., Histidine).



It's just a simple ratio!

Classification: The Dietary Need

Essential



Must eat these!

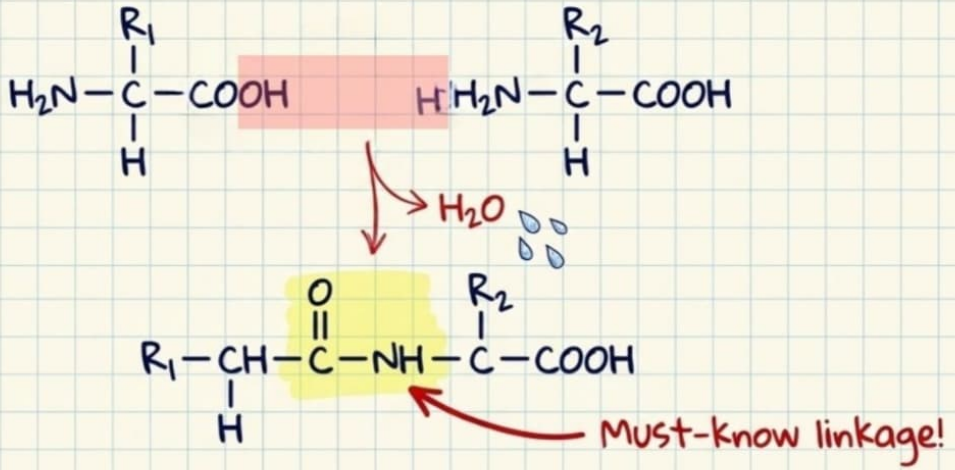
Cannot be synthesised by the body.
MUST be obtained through diet.
(e.g., Valine, Leucine).

Non-essential

Body makes these.

Our bodies can synthesise these naturally.
(e.g., Glycine, Alanine).

Connecting the Blocks: The Peptide Bond



- Mechanism: The carboxyl group of one molecule combines with the amino group of another.
- Result: Elimination of a water molecule creates a strong amide linkage (-CO-NH-), forming dipeptides, polypeptides, and eventually full Proteins!

NotebookLM

Protein Classification: The Shape of Life

Fibrous Proteins: Polypeptide chains run parallel. Held firmly together by by hydrogen and disulphide bonds to form a fibre-like structure.

Globular Proteins: Polypeptide chains coil around themselves to give a compacted, spherical shape.

Rule of Biology: Structure = Function.

A4

Fibrous vs. Globular (The Cheat Sheet)

Feature	Fibrous	Globular
Shape	Thread-like/Parallel	Spherical/Coiled
Solubility	Insoluble in water	Soluble in water
Examples	Keratin (hair, wool), Myosin (muscles)	Insulin, Albumins

High-yield exam table! Memorise this!

Key Difference!

Summary: The Chemistry of Life

Biomolecules

Carbohydrates

- Give us ENERGY
- 6 Clues decoded the Open Chain
- Plot Twist: Cyclic Pyranose Ring.

Proteins

- Give us STRUCTURE
- Built from Amino Acids (Essential/Non-essential)
- Fold into Fibrous & Globular shapes.

Done!

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